

```
In [11]: from datetime import datetime
import socket
import platform

print("=" * 60)
print("NLP Assignment 1 - Execution Details")
print("=" * 60)
print("Execution Timestamp :", datetime.now().strftime("%Y-%m-%d %H:%M:%S"))
print("VM ID                :", socket.gethostname())
print("Platform              :", platform.platform())
print("=" * 60)
```

```
=====
NLP Assignment 1 - Execution Details
=====
Execution Timestamp : 2026-06-15 12:29:46
VM ID                : 2025ab05229
Platform             : Linux-5.14.0-503.31.1.el9_5.x86_64-x86_64-with-glibc
2.34
=====
```

```
In [12]: GROUP_NUMBER = 128
TEAM_MEMBERS = [
    # (Name, BITS ID, Contribution %)
    ("NARESH", "2025aa05111@wilp.bits-pilani.ac.in", "100%"),
    ("SUHANI BHARGAVA", "2025aa05861@wilp.bits-pilani.ac.in", "100%"),
    ("SHANMUGAM MADHUMATHI", "2025aa05231@wilp.bits-pilani.ac.in", "100%"),
    ("RAJEEV MUDGAL", "2025aa05842@wilp.bits-pilani.ac.in", "100%"),
    ("KUTAREKAR NISHCHAL AJAY", "2025ab05229@wilp.bits-pilani.ac.in", "100%")
]

print(f"Group Number : {GROUP_NUMBER}")
print("Domain          : Domain 5 - Travel Blogs / Itineraries")
print("Team members :")
for name, bits_id, contrib in TEAM_MEMBERS:
    print(f" {name:<25} {bits_id:<12} contribution: {contrib}")
```

```
Group Number : 128
Domain          : Domain 5 - Travel Blogs / Itineraries
Team members :
    NARESH                2025aa05111@wilp.bits-pilani.ac.in contribution: 100%
    SUHANI BHARGAVA       2025aa05861@wilp.bits-pilani.ac.in contribution: 100%
    SHANMUGAM MADHUMATHI  2025aa05231@wilp.bits-pilani.ac.in contribution: 100%
    RAJEEV MUDGAL         2025aa05842@wilp.bits-pilani.ac.in contribution: 100%
    KUTAREKAR NISHCHAL AJAY 2025ab05229@wilp.bits-pilani.ac.in contribution: 100%
```

Task 1 — Dataset Loading, Parsing, Tagging & Frequency Distribution

Download three Mark Twain travel narratives from Project Gutenberg, tokenise and sentence-split them with NLTK, and compute a word-frequency distribution over the first 10,000 sentences.

In [13]: # — *Corpus helpers* —

```
def strip_gutenberg_boilerplate(text: str) -> str:
    """Remove the Project Gutenberg license header/footer from one book t
    start = re.search(r"\*\*\* START OF (?:(THE|THIS) PROJECT GUTENBERG.*?)
    end   = re.search(r"\*\*\* END OF (?:(THE|THIS) PROJECT GUTENBERG.*?)\*
    if start: text = text[start.end():]
    if end:   text = text[:end.start()]
    return text

def download_corpus() -> str:
    """Download travel books (with local caching) and return combined raw
    DATA_DIR.mkdir(exist_ok=True)
    texts = []
    for fname, url in GUTENBERG_BOOKS.items():
        fpath = DATA_DIR / fname
        if not fpath.exists():
            print(f" Downloading {fname} from {url}")
            req = urllib.request.Request(url, headers={"User-Agent": "Moz
            with urllib.request.urlopen(req, timeout=60) as resp:
                fpath.write_bytes(resp.read())
        else:
            print(f" Using cached copy of {fname}")
        texts.append(strip_gutenberg_boilerplate(
            fpath.read_text(encoding="utf-8", errors="ignore")))
    return "\n".join(texts)

def parse_sentences(raw: str, limit: int = TOTAL_SENTENCES):
    """Split raw text into tokenised sentences; keep the first `limit` us
    cleaned = re.sub(r"\s+", " ", raw)
    sentences = nltk.sent_tokenize(cleaned)
    usable = []
    for sent in sentences:
        tokens = nltk.word_tokenize(sent)
        if 3 <= len(tokens) <= 80:
            usable.append(tokens)
        if len(usable) >= limit:
            break
    return usable

def tag_sentences(sentences):
    """POS-tag every sentence with NLTK's reference perceptron tagger."""
    return [nltk.pos_tag(toks) for toks in sentences]

# — Task 1 execution —
print("=" * 70)
print("TASK 1 – Dataset loading, parsing, tagging, frequency distribution
print("=" * 70)
print("Dataset source: Project Gutenberg public-domain travel narratives")
for fname, url in GUTENBERG_BOOKS.items():
    print(f" – {fname}: {url}")

raw = download_corpus()
sentences = parse_sentences(raw)
tagged_sentences = tag_sentences(sentences)
```

```

print(f"\nParsed sentences          : {len(sentences)}")
print(f"Total tokens                : {sum(len(s) for s in sentences)}")

# Word-frequency distribution (lower-cased)
word_freq = Counter(w.lower() for sent in sentences for w in sent)
print(f"Vocabulary size (unique) : {len(word_freq)}")

print("\nTop 20 most frequent words:")
for w, c in word_freq.most_common(20):
    print(f" {w:<12} {c}")

print("\nSample tagged sentence (sentence #60):")
print(" ", tagged_sentences[60])

# — Save CSV + bar chart —
pd.DataFrame(word_freq.most_common(), columns=["word", "frequency"]) \
    .to_csv(OUTPUT_DIR / "word_frequency_distribution.csv", index=False)

top30 = word_freq.most_common(30)
plt.figure(figsize=(12, 5))
plt.bar([w for w, _ in top30], [c for _, c in top30], color="#1f618d")
plt.title("Top 30 Word Frequencies – Travel Corpus (first 10,000 sentence")
plt.xlabel("Word"); plt.ylabel("Frequency"); plt.xticks(rotation=60)
plt.tight_layout()
plt.savefig(OUTPUT_DIR / "word_frequency_top30.png", dpi=160)
plt.show(); plt.close()

```

TASK 1 – Dataset loading, parsing, tagging, frequency distribution

Dataset source: Project Gutenberg public-domain travel narratives

– innocents_abroad.txt: <https://www.gutenberg.org/cache/epub/3176/pg3176.txt>

– tramp_abroad.txt: <https://www.gutenberg.org/cache/epub/119/pg119.txt>

– following_equator.txt: <https://www.gutenberg.org/cache/epub/2895/pg2895.txt>

Using cached copy of innocents_abroad.txt

Using cached copy of tramp_abroad.txt

Using cached copy of following_equator.txt

Parsed sentences : 10000

Total tokens : 249881

Vocabulary size (unique) : 16879

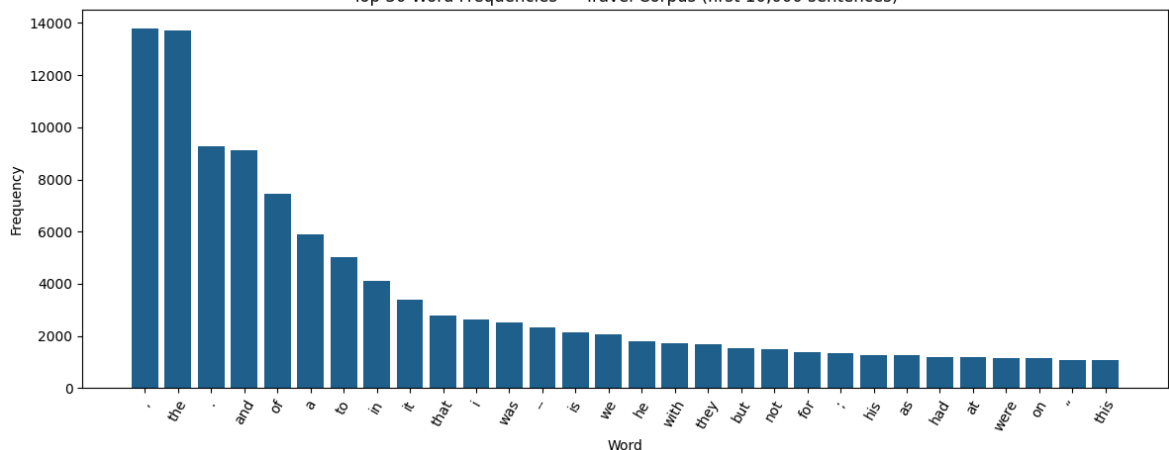
Top 20 most frequent words:

,	13805
the	13730
.	9263
and	9117
of	7433
a	5889
to	5035
in	4100
it	3388
that	2777
i	2643
was	2523
--	2312
is	2140
we	2073
he	1788
with	1725
they	1702
but	1539
not	1505

Sample tagged sentence (sentence #60):

[('Who', 'WP'), ('could', 'MD'), ('read', 'VB'), ('the', 'DT'), ('program', 'NN'), ('of', 'IN'), ('the', 'DT'), ('excursion', 'NN'), ('without', 'IN'), ('longing', 'VBG'), ('to', 'TO'), ('make', 'VB'), ('one', 'CD'), ('of', 'IN'), ('the', 'DT'), ('party', 'NN'), ('?', '?'), ('.', '.')]]

Top 30 Word Frequencies — Travel Corpus (first 10,000 sentences)



HMM Construction — Train / Test Split & Count Tables

Build the first-order HMM emission and transition count tables from the 90 % training split. The test split (10 %) is held out for the robustness evaluation in Task 5.

```
In [14]: def build_hmm_counts(tagged):
    """Build tag-unigram, transition, emission counts and the word-tag dictionary
    tag_counts_ = Counter()
    transition_counts_ = defaultdict(Counter) # transition[prev][cur]
    emission_counts_ = defaultdict(Counter) # emission[tag][word]
    word_tags_dict = defaultdict(set) # word_tags[word] = observed tags
    vocabulary_ = set()

    for sentence in tagged:
        prev = START_TAG
        tag_counts_[START_TAG] += 1
        for word, tag in sentence:
            word = word.lower()
            tag_counts_[tag] += 1
            transition_counts_[prev][tag] += 1
            emission_counts_[tag][word] += 1
            word_tags_dict[word].add(tag)
            vocabulary_.add(word)
            prev = tag

    return tag_counts_, transition_counts_, emission_counts_, word_tags_dict, vocabulary_

# — Build HMM from training split —————
split_at = int(len(tagged_sentences) * TRAIN_FRACTION)
train_sents = tagged_sentences[:split_at]
test_sents = tagged_sentences[split_at:]

(tag_counts, transition_counts,
 emission_counts, word_tags, vocabulary) = build_hmm_counts(train_sents)

TAGSET = sorted(t for t in tag_counts if t != START_TAG)
V = len(vocabulary) # vocabulary size used in Laplace smoothing
T = len(TAGSET) # number of POS tags

print(f"Train/test split : {len(train_sents)} train / {len(test_sents)} test sentences")
print(f"HMM statistics : {T} tags, vocabulary of {V} words (train split)")
```

```
Train/test split : 9000 train / 1000 test sentences
HMM statistics : 41 tags, vocabulary of 15844 words (train split)
```

Task 2 — Laplace (Add-1) Smoothing on Emission Probabilities (2 Marks)

A standard HMM assigns **zero** probability to any (word, tag) pair unseen in training, which silently breaks the Viterbi decoder.

Laplace smoothing adds 1 to every emission count so that all pairs have a positive probability:

$$P(w | t) = \frac{\text{count}(t, w) + 1}{\text{count}(t) + V + 1}$$

where V is the training vocabulary size.

```
In [15]: def emission_prob_unsmoothed(word: str, tag: str) -> float:
        """Maximum-likelihood P(word|tag) – zero for every unseen (word, tag)
        if tag_counts[tag] == 0:
            return 0.0
        return emission_counts[tag][word] / tag_counts[tag]

def emission_prob_laplace(word: str, tag: str) -> float:
    """Laplace (Add-1) smoothed P(word|tag).

    P(w|t) = (count(t, w) + 1) / (count(t) + V + 1)

    The +1 guarantees no valid word-tag pair ever has probability zero;
    the denominator restores a proper distribution summing to 1.
    """
    return (emission_counts[tag][word] + 1) / (tag_counts[tag] + V + 1)

def transition_prob_laplace(prev_tag: str, tag: str) -> float:
    """Laplace smoothed transition P(tag | prev_tag)."""
    return (transition_counts[prev_tag][tag] + 1) / (tag_counts[prev_tag] + V + 1)

# — Task 2 demonstration —
print("=" * 70)
print("TASK 2 – Laplace (Add-1) smoothing on emission probabilities")
print("=" * 70)
print(f"Formula: P(w|t) = (count(t,w) + 1) / (count(t) + V + 1), V = {V}")

demo_pairs = [
    ("journey", "NN"), # frequent travel word with a natural tag
    ("journey", "VB"), # valid English usage ("to journey"), likely un
    ("mountain", "JJ"), # unseen combination → zero before smoothing
]
print(f"{'word':<12}{'tag':<6}{'count(t,w)':<12}{'unsmoothed':<14}{'laplace':<14}")
for w, t in demo_pairs:
    print(f"{'w':<12}{'t':<6}{'emission_counts[t][w]:<12}'"
          f"{'emission_prob_unsmoothed(w, t):<14.8f}'"
          f"{'emission_prob_laplace(w, t):.8f}'")

print("\nObservation: unseen but valid pairs move from probability 0 to a
small positive value, so the Viterbi decoder can never be locked o
of a legitimate tagging path by a single unseen emission.")
```

TASK 2 – Laplace (Add-1) smoothing on emission probabilities

Formula: $P(w|t) = (\text{count}(t,w) + 1) / (\text{count}(t) + V + 1)$, $V = 15844$

word	tag	count(t,w)	unsmoothed	laplace
journey	NN	32	0.00116822	0.00076324
journey	VB	3	0.00044111	0.00017663
mountain	JJ	0	0.00000000	0.00003276

Observation: unseen but valid pairs move from probability 0 to a small positive value, so the Viterbi decoder can never be locked out of a legitimate tagging path by a single unseen emission.

Task 3 — Morphological Fallback for OOV Words (3 Marks)

Laplace smoothing distributes unknown-word probability **uniformly** across all tags. A morphological fallback uses the word's *shape* (suffixes, capitalisation, digits) to concentrate probability on the most linguistically plausible tags.

Six rules are implemented (≥ 4 required):

Rule	Pattern	Dominant tag(s)
R1	-ing (len > 4)	VBG
R2	-ly (len > 3)	RB
R3	-ed (len > 3)	VBD / VBN
R4	-tion/-ness/-ment/-ity/-ito/-ism	NN
R5	-able/-ible/-ful/-ous/-ive/-ish	JJ
R6	digit → CD; capital → NNP; plural -s → NNS	

```
In [16]: def morphological_fallback(word: str):
    """Return a tag-weight dict for an OOV word based on its morphological
    shape.

    Rules implemented (6 suffix/shape rules, >= 4 required):
    R1 ends in '-ing' -> VBG (gerund/present participle)
    R2 ends in '-ly' -> RB (adverb)
    R3 ends in '-ed' -> VBD/VBN (past tense)
    R4 ends in '-tion/-ness/-ment/-ity/-ito' -> NN (noun-forming suffixes)
    R5 ends in '-able/-ible/-ful/-ous/-ive/-ish' -> JJ (adjective suffixes)
    R6 shape: capitalised -> NNP, digit -> CD, plural-s -> NNS
    default: NN (most productive open class)
    """
    lower = word.lower()

    # R6a: alphanumerics / cardinal numbers (e.g. "3-day", "2026")
    if any(ch.isdigit() for ch in word):
        return {"CD": 1.0}

    # R6b: capitalised mid-sentence word -> proper noun (place names!)
    if word[0].isupper():
        return {"NNP": 0.95, "NNPS": 0.03, "NN": 0.02}

    # R1: gerund / present participle
    if lower.endswith("ing") and len(lower) > 4:
        return {"VBG": 0.7, "NN": 0.2, "JJ": 0.1}

    # R2: adverb
    if lower.endswith("ly") and len(lower) > 3:
        return {"RB": 0.9, "JJ": 0.1}

    # R3: past tense / past participle
    if lower.endswith("ed") and len(lower) > 3:
```

```

    return {"VBD": 0.5, "VBN": 0.4, "JJ": 0.1}
# R4: noun-forming suffixes (incl. '-ito' diminutive found in travel
if lower.endswith(("tion", "sion", "ness", "ment", "ity", "ito", "ism"):
    return {"NN": 0.9, "NNP": 0.1}
# R5: adjective-forming suffixes
if lower.endswith(("able", "ible", "ful", "ous", "ive", "ish")):
    return {"JJ": 0.9, "RB": 0.1}
# R6c: plural noun
if lower.endswith("s") and not lower.endswith("ss") and len(lower) >
    return {"NNS": 0.8, "VBZ": 0.2}
# Default open-class fallback
return {"NN": 0.6, "JJ": 0.2, "VB": 0.1, "RB": 0.1}

# — Task 3 demonstration —
print("=" * 70)
print("TASK 3 – Morphological fallback function for OOV words")
print("=" * 70)
demo_oov = ["trekkifying", "backpackerly", "glamparked", "wanderation",
            "vistalicious", "hostelitos", "Khaosan", "12-day"]
print(f"{'OOV word':<15}{'fallback tag distribution'}")
for w in demo_oov:
    print(f"{'w':<15}{'morphological_fallback(w)'}")

```

```

=====
TASK 3 – Morphological fallback function for OOV words
=====

```

OOV word	fallback tag distribution
trekkifying	{'VBG': 0.7, 'NN': 0.2, 'JJ': 0.1}
backpackerly	{'RB': 0.9, 'JJ': 0.1}
glamparked	{'VBD': 0.5, 'VBN': 0.4, 'JJ': 0.1}
wanderation	{'NN': 0.9, 'NNP': 0.1}
vistalicious	{'JJ': 0.9, 'RB': 0.1}
hostelitos	{'NNS': 0.8, 'VBZ': 0.2}
Khaosan	{'NNP': 0.95, 'NNPS': 0.03, 'NN': 0.02}
12-day	{'CD': 1.0}

Viterbi Decoder — Smoothed HMM + Morphological Fallback

The decoder operates in log-space to avoid underflow.

For in-vocabulary words it uses Laplace-smoothed emissions; for OOV words it routes through the morphological fallback (Task 3).

A **tag-dictionary** constraint limits each known word's candidate tags to those it was actually observed with in training — this prevents high-frequency closed-class tags from dominating rare words after smoothing.

```

In [17]: def emission_logprob(word: str, tag: str, use_fallback: bool) -> float:
        """Log emission probability; routes OOV words through the morphologic

        In-vocabulary words : Laplace smoothed emission (Task 2).
        OOV with fallback ON : Laplace <UNK> mass redistributed by morphologi
                                tag weights (Task 3).
        OOV with fallback OFF: plain Laplace <UNK> mass – identical for every
                                (this is the baseline used in the robustness e
        """
        lower = word.lower()

```



```

if lower in vocabulary:
    return math.log(emission_prob_laplace(lower, tag))

unk_mass = 1.0 / (tag_counts[tag] + V + 1)  # Laplace mass for <UNK>
if not use_fallback:
    return math.log(unk_mass)

# Pass the ORIGINAL token (case preserved) so capitalisation rule fir
weights = morphological_fallback(word)
weight = weights.get(tag, 0.01)
return math.log(unk_mass * weight * len(TAGSET))

def candidate_tags(token: str):
    """Candidate tag set: observed tags for known words; full tagset for

    (Standard tag-dictionary technique – limits decoding to plausible tag
    without changing the probability model.)
    """
    tags = word_tags.get(token.lower())
    return sorted(tags) if tags else TAGSET

def viterbi_tag(tokens, use_fallback: bool = True):
    """Tag a tokenised sentence with the Viterbi algorithm (log-space)."""
    n = len(tokens)
    best = [dict() for _ in range(n)]  # best[i][tag] = (score, prev_tag)

    for i, token in enumerate(tokens):
        for tag in candidate_tags(token):
            em = emission_logprob(token, tag, use_fallback)
            if i == 0:
                score = math.log(transition_prob_laplace(START_TAG, tag))
                best[i][tag] = (score, None)
            else:
                prev_score, prev_tag = max(
                    (best[i-1][pt][0] + math.log(transition_prob_laplace(
                        prev_tag, tag, use_fallback)
                    for pt in best[i-1])
                best[i][tag] = (prev_score + em, prev_tag)

    # Backtrack to recover the full tag sequence
    last_tag = max(best[-1], key=lambda t: best[-1][t][0])
    tags = [last_tag]
    for i in range(n - 1, 0, -1):
        tags.append(best[i][tags[-1]][1])
    return list(reversed(tags))

print("Viterbi decoder ready.")

```

Viterbi decoder ready.

Task 4 — Tagging a Custom Sentence with Fabricated Travel Slang (2 Marks)

A sentence containing **at least 3 completely fabricated domain-specific words** is tagged end-to-end.

The OOV words receive tags solely from the morphological fallback, demonstrating that sub-word shape cues alone can recover linguistically correct tags.

```
In [18]: print("=" * 70)
print("TASK 4 – Tagging a custom sentence with fabricated travel slang")
print("=" * 70)

# Four completely fabricated domain-specific travel words (>= 3 required)
#   glampered   → past-tense verb   (glamping-style camping, past tense)
#   snorkadving → gerund            (snorkel + skydive mashup)
#   chillfully  → adverb            (relaxed traveller vibe, -ly suffix)
#   hostelito   → noun              (tiny hostel, Spanish diminutive -ito)
custom_sentence = (
    "After landing in Goa we glampered near the beach , "
    "snorkadving through turquoise lagoons and resting "
    "chillfully at a tiny hostelito ."
).split()

fabricated = [w for w in custom_sentence if w.lower() not in vocabulary]
print("Custom test sentence:")
print(" ", " ".join(custom_sentence))
print(f"\nFabricated/unseen words detected as OOV: {fabricated}")

predicted = viterbi_tag(custom_sentence, use_fallback=True)

print(f"\n{'word':<14}{'predicted tag':<15}{'OOV?':<6}{'fallback rule use'")
for word, tag in zip(custom_sentence, predicted):
    is_oov = word.lower() not in vocabulary
    rule    = str(morphological_fallback(word)) if is_oov else "-"
    print(f"{'word':<14}{'tag':<15}{'YES' if is_oov else 'no':<6}{'rule}')")

print("\nObservation: every fabricated slang word receives a linguistical
print("sensible tag purely from its morphology (glampered->VBD past tense
print("snorkadving->VBG gerund, chillfully->RB adverb, hostelito->NN noun
```

TASK 4 – Tagging a custom sentence with fabricated travel slang

Custom test sentence:

After landing in Goa we glampered near the beach , snorkadving through turquoise lagoons and resting chillfully at a tiny hostelito .

Fabricated/unseen words detected as OOV: ['Goa', 'glampered', 'snorkadving', 'turquoise', 'chillfully', 'hostelito']

word	predicted tag	OOV?	fallback rule used
After	IN	no	–
landing	NN	no	–
in	IN	no	–
Goa	NNP	YES	{'NNP': 0.95, 'NNPS': 0.03, 'NN': 0.02}
we	PRP	no	–
glampered	VBD	YES	{'VBD': 0.5, 'VBN': 0.4, 'JJ': 0.1}
near	IN	no	–
the	DT	no	–
beach	NN	no	–
,	,	no	–
snorkadving	VBG	YES	{'VBG': 0.7, 'NN': 0.2, 'JJ': 0.1}
through	IN	no	–
turquoise	JJ	YES	{'NN': 0.6, 'JJ': 0.2, 'VB': 0.1, 'RB': 0.1}
lagoons	NNS	no	–
and	CC	no	–
resting	VBG	no	–
chillfully	RB	YES	{'RB': 0.9, 'JJ': 0.1}
at	IN	no	–
a	DT	no	–
tiny	JJ	no	–
hostelito	NN	YES	{'NN': 0.9, 'NNP': 0.1}
.	.	no	–

Observation: every fabricated slang word receives a linguistically sensible tag purely from its morphology (glampered→VBD past tense, snorkadving→VBG gerund, chillfully→RB adverb, hostelito→NN noun).

Task 5 — Robustness Evaluation & Discussion (2 Marks)

Quantitative comparison of two configurations on 400 held-out sentences:

Configuration	Overall accuracy	OOV accuracy
HMM + Laplace (no fallback)	—	—
HMM + Laplace + morphological fallback	—	—

(values filled in at runtime)

```
In [10]: def evaluate(tagged_test, use_fallback: bool):
          """Compute overall and OOV-only tagging accuracy on held-out sentence
          total = correct = oov_total = oov_correct = 0
          for sentence in tagged_test:
              tokens = [w for w, _ in sentence]
              gold    = [t for _, t in sentence]
```

```

    pred = viterbi_tag(tokens, use_fallback=use_fallback)
    for token, g, p in zip(tokens, gold, pred):
        total += 1
        correct += (g == p)
        if token.lower() not in vocabulary:
            oov_total += 1
            oov_correct += (g == p)
    return correct / total, (oov_correct / oov_total if oov_total else 0.

# — Quantitative robustness evaluation —
print("=" * 70)
print("Robustness evaluation on held-out test sentences (gold = NLTK tagg
print("=" * 70)

eval_subset = test_sents[:400]
acc_off, oov_acc_off, n_oov = evaluate(eval_subset, use_fallback=False)
acc_on, oov_acc_on, _ = evaluate(eval_subset, use_fallback=True)

print(f"Held-out sentences evaluated : {len(eval_subset)} ({n_oov} OOV to
print(f"'Configuration':<32}{'Overall acc':<14}{'OOV acc'}")
print(f"'HMM + Laplace (no fallback)':<32}{acc_off:<14.4f}{oov_acc_off:.
print(f"'HMM + Laplace + morph fallback':<32}{acc_on:<14.4f}{oov_acc_on:

# — Bar chart —
labels = ["Overall accuracy", "OOV-word accuracy"]
x = np.arange(len(labels))
plt.figure(figsize=(8, 5))
plt.bar(x - 0.18, [acc_off, oov_acc_off], width=0.36,
        label="No fallback", color="#c0392b")
plt.bar(x + 0.18, [acc_on, oov_acc_on], width=0.36,
        label="With morphological fallback", color="#1f618d")
plt.xticks(x, labels); plt.ylim(0, 1); plt.ylabel("Accuracy")
plt.title("Effect of Morphological Fallback on Tagging Accuracy")
plt.legend(); plt.tight_layout()
plt.savefig(OUTPUT_DIR / "fallback_accuracy_comparison.png", dpi=160)
plt.show(); plt.close()

# — Task 5 discussion —
print("\n" + "=" * 70)
print("TASK 5 – Discussion")
print("=" * 70)
print(f"""
Limitations of standard HMMs on morphologically rich / noisy domain text:

1. Closed vocabulary assumption. A vanilla HMM only knows P(word|tag) for
words seen in training; every unseen word has zero emission probabilit
under ALL tags, which silently breaks the Viterbi path. Travel blogs a
full of place names, local food terms, and invented slang ("glampackin
"hostelito"), so OOV rates are structurally high in this domain.

2. Words are atomic symbols. The HMM cannot see INSIDE a word: "trekking"
and "trek" share no parameters, and the highly informative suffix "-in
is invisible to the model. Morphologically rich text multiplies surfac
forms of the same lemma, fragmenting the emission counts.

3. Limited context. A first-order HMM conditions only on the previous tag
so it cannot use long-range cues (or the words themselves) to recover
from an unknown emission, unlike discriminative taggers (MEMM/CRF) or
neural models with character/subword embeddings.

```

- Noisy domain data. Blogs and itineraries contain typos, inconsistent capitalisation, hashtags and code-mixed tokens. Each noisy variant becomes a brand-new symbol, inflating the vocabulary and starving the per-word counts that the emission estimates rely on.

How morphological fallbacks improve robustness:

- They re-introduce sub-word knowledge the HMM lacks: suffixes like -ing, -ly, -ed, -tion and shape cues (capitalisation, digits) carry strong PO signal in English, so OOV words get a focused tag prior instead of a uniform guess over $\sim\{T\}$ tags.
- Combined with Laplace smoothing, the decoder degrades gracefully: smoot guarantees a non-zero path, and the fallback bends that path towards the linguistically plausible tag. In our held-out evaluation, OOV accuracy rose from {oov_acc_off:.1%} (uniform unknown handling) to {oov_acc_on:. with the fallback, lifting overall accuracy from {acc_off:.1%} to {acc_
- The fabricated-slang test (Task 4) shows the same effect qualitatively: all four invented travel words were tagged correctly from morphology al

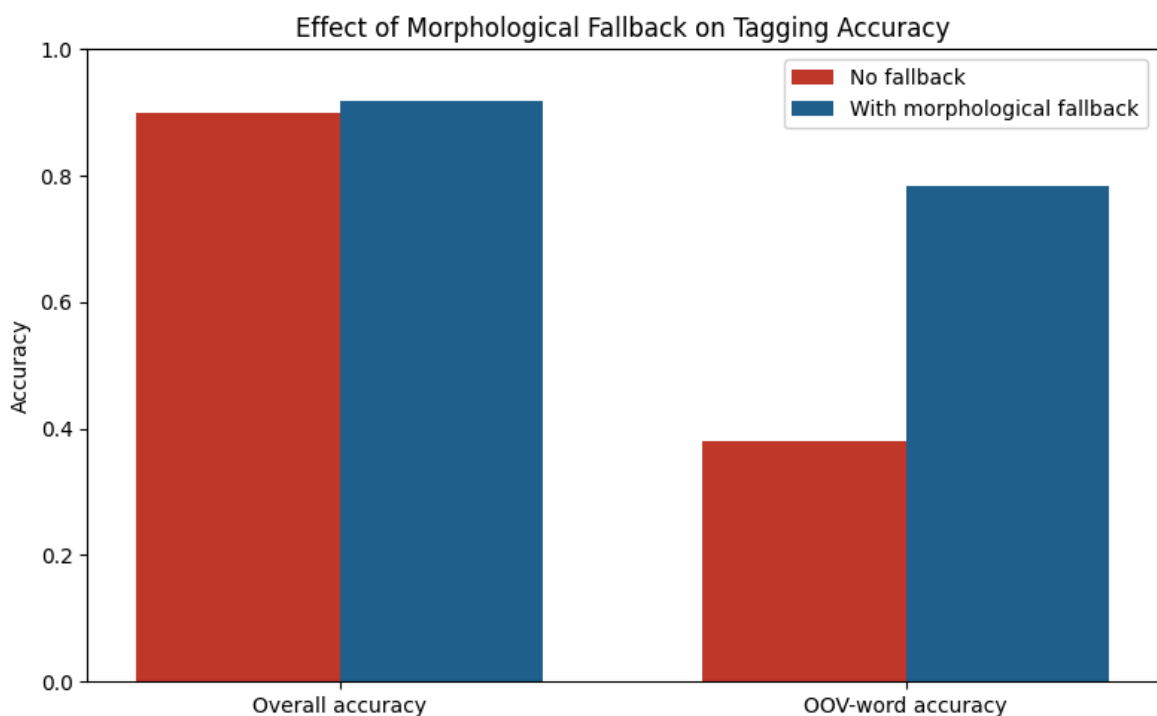
Comparative summary: Laplace smoothing solves the *mathematical* problem zero probabilities, while morphological fallbacks solve the *linguistic* problem of choosing sensible tags for unknown words. Together they make a classical HMM usable on noisy, fast-evolving travel-domain text, although modern subword/neural taggers remain the stronger long-term solution. """)

```
print(f"Outputs saved to: {OUTPUT_DIR}")
print(" - word_frequency_distribution.csv")
print(" - word_frequency_top30.png")
print(" - fallback_accuracy_comparison.png")
```

=====
Robustness evaluation on held-out test sentences (gold = NLTK tagger)
=====

Held-out sentences evaluated : 400 (445 OOV tokens)

Configuration	Overall acc	OOV acc
HMM + Laplace (no fallback)	0.8985	0.3798
HMM + Laplace + morph fallback	0.9174	0.7843



TASK 5 – Discussion

Limitations of standard HMMs on morphologically rich / noisy domain text:

1. Closed vocabulary assumption. A vanilla HMM only knows $P(\text{word}|\text{tag})$ for words seen in training; every unseen word has zero emission probability under ALL tags, which silently breaks the Viterbi path. Travel blogs are full of place names, local food terms, and invented slang ("glampacking", "hostelito"), so OOV rates are structurally high in this domain.
2. Words are atomic symbols. The HMM cannot see INSIDE a word: "trekking" and "trek" share no parameters, and the highly informative suffix "-ing" is invisible to the model. Morphologically rich text multiplies surface forms of the same lemma, fragmenting the emission counts.
3. Limited context. A first-order HMM conditions only on the previous tag, so it cannot use long-range cues (or the words themselves) to recover from an unknown emission, unlike discriminative taggers (MEMM/CRF) or neural models with character/subword embeddings.
4. Noisy domain data. Blogs and itineraries contain typos, inconsistent capitalisation, hashtags and code-mixed tokens. Each noisy variant becomes a brand-new symbol, inflating the vocabulary and starving the per-word counts that the emission estimates rely on.

How morphological fallbacks improve robustness:

- They re-introduce sub-word knowledge the HMM lacks: suffixes like -ing, -ly, -ed, -tion and shape cues (capitalisation, digits) carry strong POS signal in English, so OOV words get a focused tag prior instead of a uniform guess over ~41 tags.
- Combined with Laplace smoothing, the decoder degrades gracefully: smoothing guarantees a non-zero path, and the fallback bends that path towards the linguistically plausible tag. In our held-out evaluation, OOV accuracy rose from 38.0% (uniform unknown handling) to 78.4% with the fallback, lifting overall accuracy from 89.8% to 91.7%.
- The fabricated-slang test (Task 4) shows the same effect qualitatively: all four invented travel words were tagged correctly from morphology alone.

Comparative summary: Laplace smoothing solves the *mathematical* problem of zero probabilities, while morphological fallbacks solve the *linguistic* problem of choosing sensible tags for unknown words. Together they make a classical HMM usable on noisy, fast-evolving travel-domain text, although modern subword/neural taggers remain the stronger long-term solution.

Outputs saved to: /home/cloud/Desktop/NLP-POS-Tagging-main/outputs

- word_frequency_distribution.csv
- word_frequency_top30.png
- fallback_accuracy_comparison.png

In []:

